

1. INTRODUCTION

One of the major challenges in designing smart systems is enabling machines to not only decipher what a person is saying but also understand how they are feeling when they say it. Spoken communication provides far more context than just words. Tone of voice, facial expressions and physical gesture all carry emotional value and often these different channels convey messages in a real-world context. A machine that can only process words will likely misinterpret the speaker's emotional state. This challenge has practical relevance in many contexts, such as conversational AI assistants, which will need to vary replies according to the user's level of frustration or satisfaction, and customer experience management systems, which will need to use sentiment metrics to identify users who are unhappy before they stop using the product. In all these cases it requires immediate attention when the system misreads the emotion.

When trying to scale this issue, the main problem is moving from scripted, two person dialogues toward unscripted, multi-party discussions. In these situations, there are many different speakers talking at once, the background is noisy, the cameras will be positioned in various locations, and each speaker may be saying something to multiple listeners at the same time. These conditions degrade the performance of approaches trained on lab-quality recordings (IEMOCAP dataset, University of Southern California), and the resulting systems must be designed to operate without one or more of the available input channels.

The paper addresses two closely related classification tasks using the same conversational video dataset. The first is emotion recognition, which refers to how to assign the emotional label out of the seven categories (anger, fear, disgust, joy, sadness, surprise, and neutrality) to a given utterance. The second is sentiment analysis with three categories (positive, negative, and neutral). Both refers to instance-level classification tasks, meaning that each utterance will be classified by itself using all modalities at once.

1. PAPER SUMMARY

The main idea of the paper is that no single communication channel is good enough to reliably tell how someone feels in real conversations. The authors developed a multimodal system that uses four types of information to solve this problem such as the text transcript, the speech audio, the facial expressions, and the video frames that show posture and surrounding. The classifier uses all of these features to make the final prediction by combining them into one representation. The practical reason for combining them is that each one has its own flaws. Text makes things clear, but it cannot show tone of facial expression and fail to capture sarcasm. Audio can pick up changes in tone and voice, but noise can get in the way. Facial expressions can show how someone feels but they might not be clear or might be partly hidden. Video gives you body language and context but it also shows you things that are not important. So, when we use all four combined the system reduces the impact of these individual problems.

2. METHODOLOGY ANALYSIS

This section talks about the methodology used in the paper.

2.1. Model Architecture

The proposed system is structured in four independent feature extraction pipelines operating in parallel and each dedicated to one modality. Each of the pipeline receives its raw input and outputs a fixed length embedding vector to create a single multimodal embedding vector through the concatenation of all four embedding vectors, which is fed into a Multi-Layer Perceptron for outputting the final class probabilities. The system can work in unimodal or multimodal fusion mode which is the primary model of operation in this research paper. And finally, feature level fusion is used over decision-level fusion to prevent the model from losing inter-model correlations, and using feature concatenation preserves the relationships between modalities, hence enabling the MLP classifier to learn cross-modal patterns during its training process.

2.1.1. Text Modality

RoBERTa is used to process text, which is a transformer model that has been built on BERT's architecture. It uses the entire transcript of each instance to produce a representation of what was said while taking into account the choice of vocabulary, syntax and local context. The model has been initially trained before emotional classification task with the input of 58 million tweets using TweetEval benchmark for twitter. This was specifically chosen as an example of social media where emotions are often present and the expression is typically informal and contains many different types of emotional language which is similar to the language model used in conversational speech.

2.1.2. Voice Modality

Wav2Vec2 analyzes the soundwaves of an audio piece without needing standard acoustic features (MFCCs). The architecture contains two components, first a convolutional encoder creates many very similar forms of how each sound waveform appeared at specific times during recording and second, a transformer-based architecture provides a long-term history of what shapes the same sounds made throughout its entire length using self-attention. The original model was previously pre-trained through a method called masked language modeling. Using this method to learn features of an audio waveform gives Wav2Vec2 enough knowledge to understand features found in speech such as pitch, rhythm and stress which are often indicators of the emotional state of a person. Thus, this modal is then fine-tuned on MELD dataset.

2.1.3. Facial Expression Modality - FacialNet

There are two stages in facial expression pipeline, speaker face isolation and feature based emotion classification. When dealing with multiple users in a multi-party video, simply extracting every user's visible face would create far too much noise, since the listener reactions and background characters do not contribute to the speaker's emotional label. The first process utilizes an active speaker detector model (TalkNet), this detector uses audio and visual cues to identify which face in each frame belongs to the currently speaking person. The second process refines the results of the first process using the InfoMap clustering process and compares them against a pre-built character face library to eliminate any remaining face expressional data that does not pertain to the speaker. The isolated frames of the speaker's face will go through InceptionResNetV1, that was previously trained on the CASIA-WebFace dataset for the face recognition task. Then, a BiLSTM with an attention mechanism looks at the sequence of face frames to figure out what emotion it is. This method doesn't work well in practice because multiple face frames in a single clip often show expressions that do not match up with each other, even though they all have the same ground-truth label.

2.1.4. Video Modality

FacialNet only looks at the speaker's face, but the video pipeline is made to capture the whole visual context, including posture, gestures, the presence of other individuals and the background elements. There are three parts of the architecture, a spatial modelling, a temporal modelling and a classifier. MobileNetV2 has utilized to get features for spatial modelling. This architecture was chosen because it is efficient and has low latency. Then, use a transformer-based method for temporal modeling that captures the relationships between frames over time. Because long-range dependencies aren't always needed for recognizing emotions, they used a local attention mechanism to focus on the most important time-based information. This module has two transformer layers, each with eight attention heads and a hidden size of 1280. Finally, the features that were taken out are sent through a neural network classifier with a hidden size of 512 to predict the emotion labels.

2.2. Dataset(s) used

All experiments were performed using the Multimodal EmotionLines Dataset (MELD), which is made up of clips from the television series Friends. MELD contains more than 1,400 dialogues and over 13,000 utterances. Each utterance has two labels, emotion and sentiment mentioned above in narrow target task.

The paper also utilized external datasets solely for the purpose of pre-training. Wav2Vec2 has undergone pre-training via different speech-emotion datasets. InceptionResNetV1 was pre-trained using CASIA-WebFace. Evaluation, however, was based only on the MELD dataset.

2.3. Training Procedure

Transfer learning is the main method used in the paper. The authors do not train models from scratch. Instead, they use models that have already trained on large datasets and then fine-tune them on MELD. This is helpful because MELD is small and starting for scratch to train deep models could easily lead to overfitting. The idea is followed by different models. RoBERTa is first trained on general language data, then adapted using TweetEval, and finally fine-tuned on MELD. Wav2Vec2 is also trained on raw speech and other emotion speech datasets before being fine-tuned on MELD. Moreover, InceptionResNet1 is trained on the CASIA-WebFace dataset first and then it is used to extract facial features. However, the only model trained from scratch is the CNN + Transformer video model. The models use the AdamW optimizer with a batch size of 16 to train. The video model trained from scratch runs for 10 training iterations, while the pre-trained models are fine-tuned for 5 epochs.

2.4. Evaluation Metrics

The paper uses a classification accuracy solely throughout the experiments.

2.5. Baseline Used for Comparison

This paper compares the performance of its four different modalities using two methods. The first method involves reporting on the performance of each modality alone. These individual modality results act as a baseline for determining if combining modalities improves performance. The second method uses three of the four modalities (removing one at a time) and comparing these against the performance of the four modalities together. This helps to show each modality's contribution to the final result of the combined modalities.

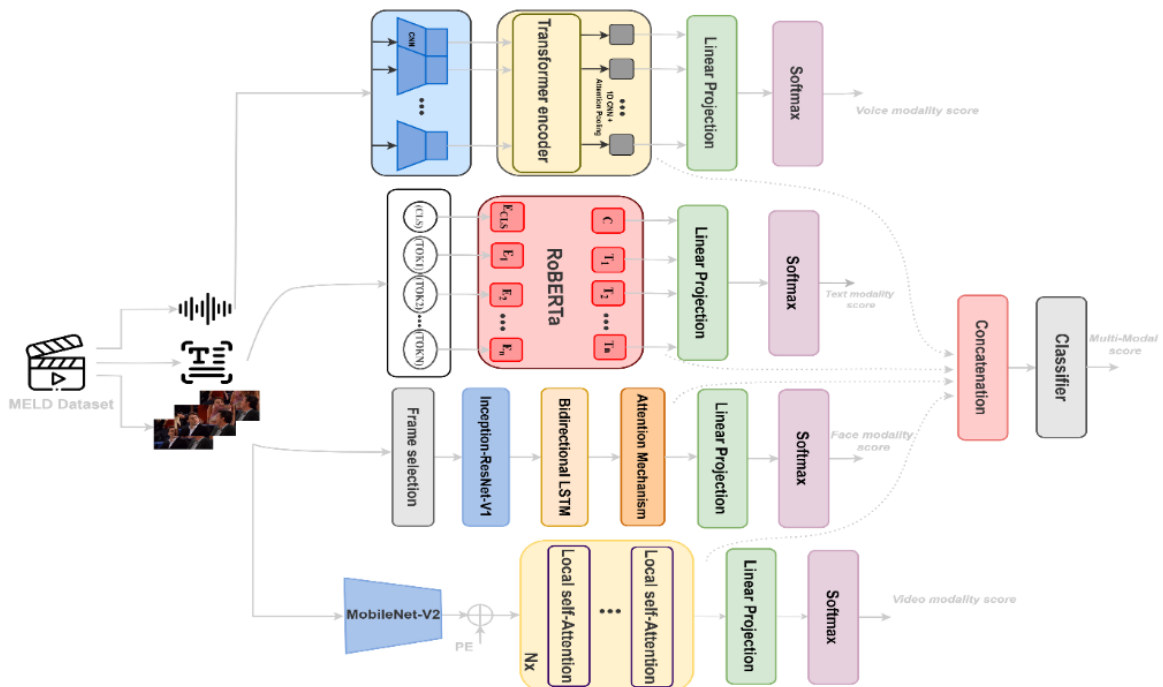


Figure 2: Proposed Methodology

3. EXPERIMENTAL SETUP

This section summarizes the results obtained from the proposed method in the paper.

3.1. Unimodal Results

Table 1. Results in percentage for emotion recognition in unimodal mode

	Text	Voice	Face	Video
Accuracy	64.34	51.49	22.61	36.14

Table 2. Results in percentage for sentiment analysis in single-modality mode

	Text	Voice	Face	Video
Accuracy	69.21	56.20	38.98	42.51

In unimodal, as compared to the audio, text performs better by 13% on average across both tasks. The visual modality performs significantly worse, with the face model showing 22.61% accuracy in determining emotion, which is just above what would be expected from randomly guessing in a seven-class problem (14%). The reason for this is because there are multiple frames per video clip, and face frames can have very different expressions even though they have the same label. The video performs marginally better (36.14%) than the face model due to the fact that video model captures a wider visual context.

3.2. Multimodal Fusion Results

Table 3. Emotion recognition in feature fusion

Modalities	Accuracy (%)
Voice + Face + Video	48.16
Text + Face + Video	65.98
Text + Face + Voice	66.25
Text + Video + Voice	66.29
Text + Face + Video + Voice	66.36

Table 4. Sentiment analysis in feature fusion

Modalities	Accuracy (%)
Voice + Face + Video	49.58
Text + Video + Voice	71.76
Text + Video + Face	71.80
Text + Face + Voice	71.84
Text + Video + Face + Voice	72.15

In multimodal, the results show that text provides the most contribution. Any combination of text with other modalities gives a similar emotion accuracy at approximately 65.98% - 66.36%. However, when text is removed, the performance dropped drastically. The voice + face + video combination received emotion and sentiment scores below 50% for both tasks. This confirms that audio and visual modalities do not capture the meaning inherent in the text, however, both provide some minor contribution when text is combined with them. On the other hand, the four modalities combined gives 66.36% for emotion and 72.15% for sentiment, higher than the text only baseline. Overall, sentiment recognition performed better than emotion recognition across all modalities.

4. CRITICAL ANALYSIS

After our critical analysis, several limitations remain:

4.1. Limitations

The main limitation of this paper is the method fusion itself. Because it uses a simple concatenation to combine features across modalities with equal weight. However, there may be noise in certain modalities in reality, thus a better alternative would be required. Moreover, the model is effective at identifying a single sentence and ignoring the previous sentences, consequently there are instances in which it misidentifies the accompanying emotion because it can be dependent upon previous sentences.

4.2. Dataset Bias or Limitations

The dataset has two issues, firstly, it was collected from a TV series, so it may not accurately reflect how people feel emotionally in the real world and secondly, some emotions are present far more frequently than others, which means the overall accuracy could disguise a lack of performance on rare emotions.

4.3. Reproducibility Concerns

The authors didn't mention the code. However, there is a GitHub repository that allows researchers for better reproducibility but still makes it difficult to exactly reproduce the system and compare its performance.

4.4. Scalability Issues

The system may be difficult to use in real time because required multiple heavy models running concurrently and does not provide any information about computational cost or inference time. Additionally, because the system was trained on English language only, there may be some difficulty generalizing on other languages.

5. FUTURE RESEARCH OPPORTUNITIES

The existing system uses only the English language, and we can expand this to multilingual emotion recognition, which will make the system more useful and applicable to real-world. Furthermore, the authors used simple feature concatenation method, which could be replaced with cross-modal attention or reliability scoring. This would let the model decide which modality should have more effect on the prediction.

6. REFERENCES

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